Association Analysis Example

The **BANK** data set contains service information for nearly 8,000 customers. There are three variables in the data set, as shown in the table below.

Name	Model Role	Measurement Level	Description
ACCOUNT	ID	Nominal	Account Number
SERVICE	Target	Nominal	Type of Service
VISIT	Sequence	Ordinal	Order of Product Purchase

The **BANK** data set has over 32,000 rows. Each row of the data set represents a customer-service combination. Therefore, a single customer can have multiple rows in the data set, and each row represents one of the products he or she owns. The median number of products per customer is three.

The 13 products are represented in the data set using the following abbreviations:

ATM	automated teller machine debit card
AUTO	automobile installment loan
CCRD	credit card
CD	certificate of deposit
CKCRD	check/debit card
CKING	checking account
HMEQLC	home equity line of credit
IRA	individual retirement account
MMDA	money market deposit account
MTG	mortgage
PLOAN	personal/consumer installment loan
SVG	saving account
TRUST	personal trust account

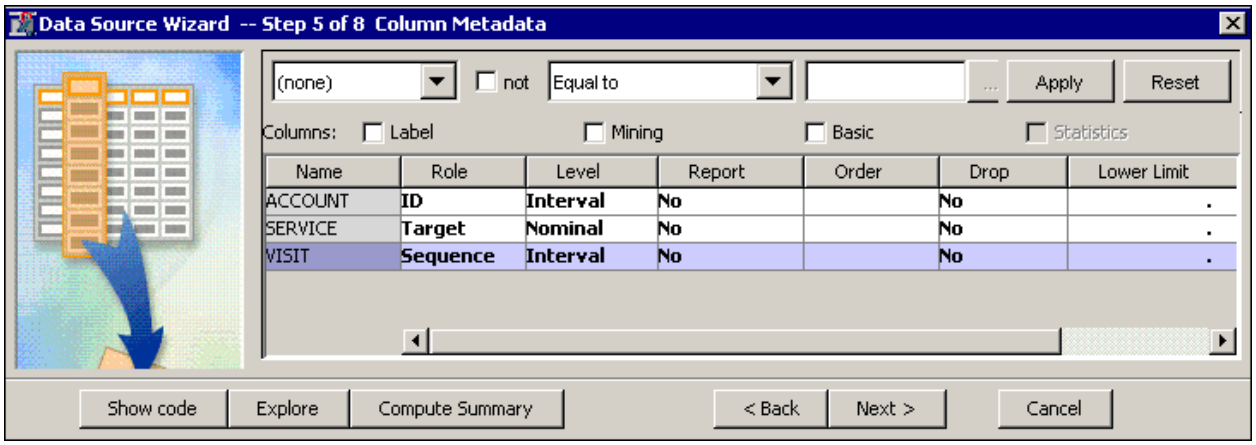
Your first task is to create a new analysis diagram and data source for the **BANK** data set.

1. Create a new diagram named Association Analysis to contain this analysis.

2. Select **Create Data Source** from the Data Sources project property.

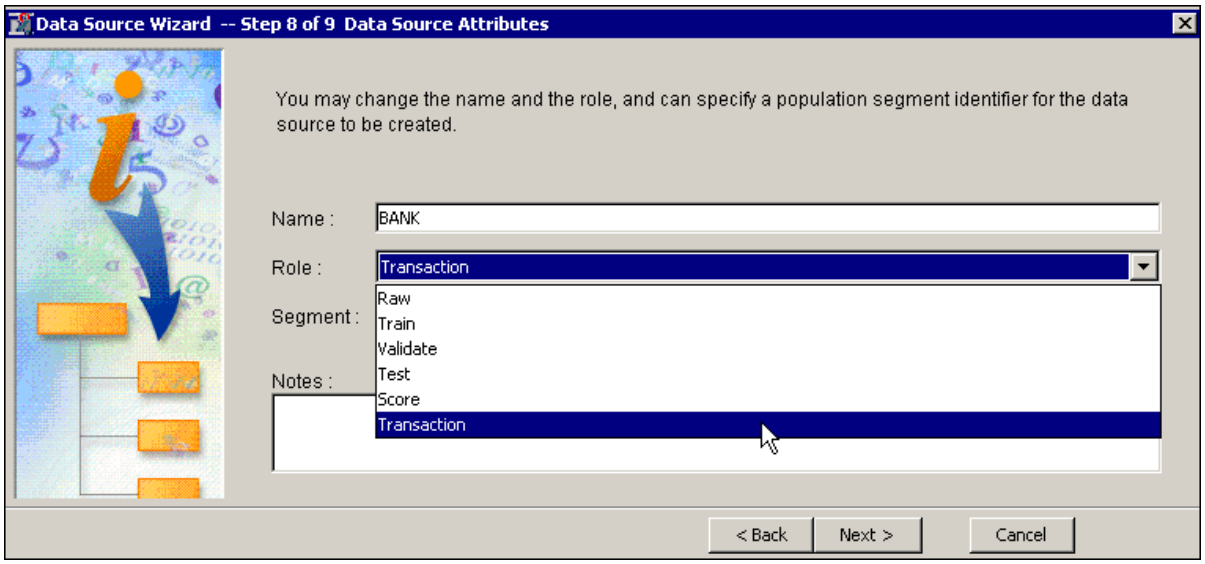
3. Select the **BANK** table from the designated library.

4. In Step 5, assign roles to the table variables as shown below.



An association analysis requires exactly one target variable and at least one ID variable. Both should have a nominal measurement level; however, a level of Interval for the ID variable is sufficient. A sequence analysis also requires a sequence variable. It usually has an ordinal measurement scale; however, in SAS Enterprise Miner the sequence variable must be assigned the level Interval.

5. For an association analysis, the data source should have a role of Transaction. In Step 9, select **Role** ⇨ **Transaction**.

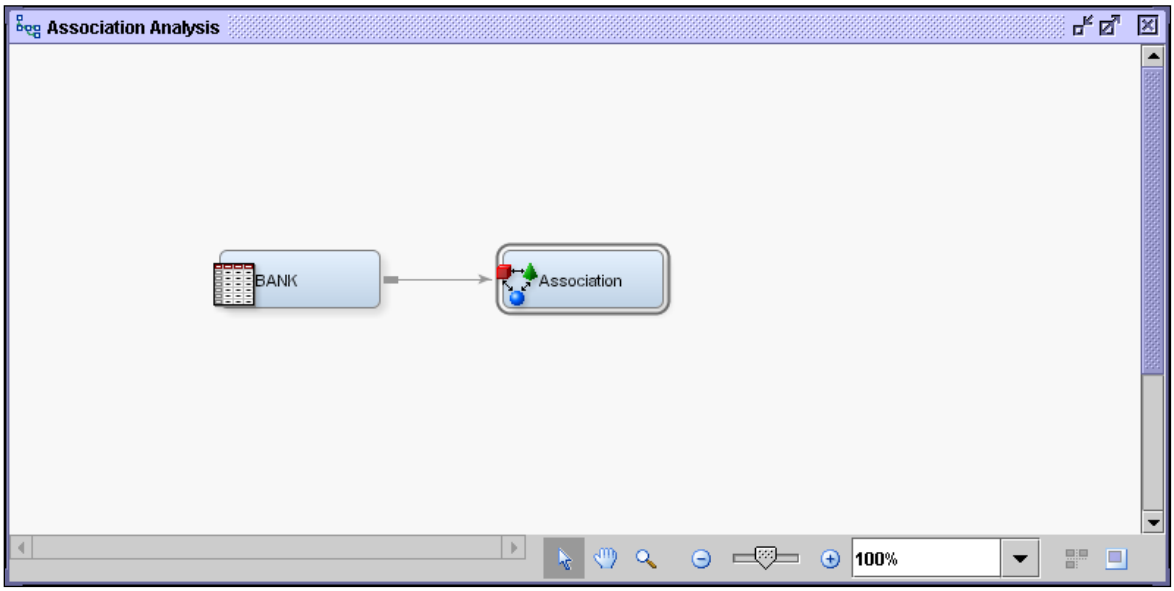


6. Select **Finish** to close the Data Source Wizard.

7. Drag a **BANK** data source into the diagram workspace.

8. Select the **Explore** tab and drag an **Association** tool into the diagram workspace.

9. Connect the **BANK** node to the **Association** node.



10. Select the **Association** node and examine its Properties panel.

Property	Value
General	
Node ID	Assoc
Imported Data	...
Exported Data	...
Notes	...
Train	
Variables	...
Maximum Number of Items to Process	100000
Rules	...
Association	
Maximum Items	4
Minimum Confidence Level	10
Support Type	Percent
Support Count	.
Support Percentage	5.0
Sequence	
Chain Count	3
Consolidate Time	0.0
Maximum Transaction Duration	.
Support Type	Percent
Support Count	.
Support Percentage	2.0
Rules	
Number to Keep	200
Sort Criterion	Default
Number to Transpose	200
Export Rule by ID	No
Recommendation	No

Options in the Properties panel include the following:

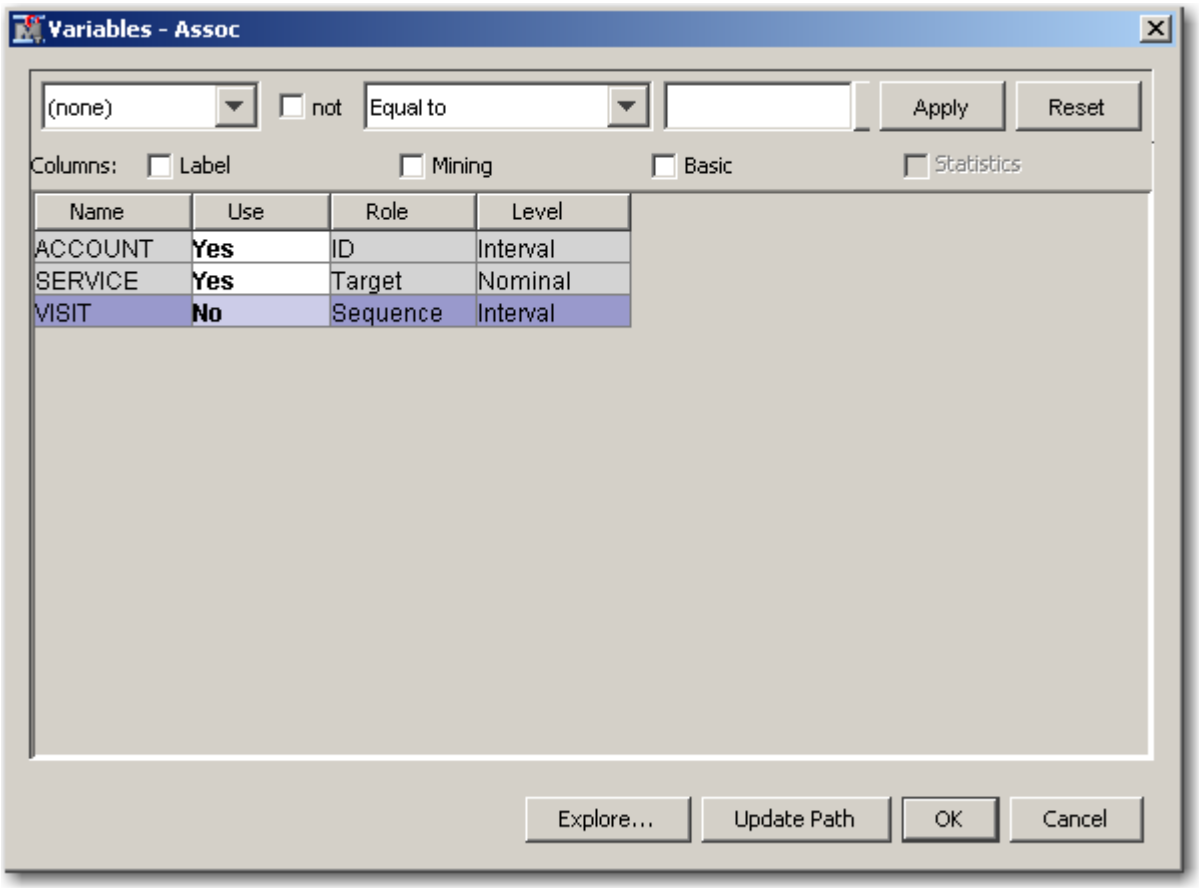
- **Minimum Confidence Level** specifies the minimum confidence level to generate a rule. The default level is **10%**.
- **Support Type** specifies whether the analysis should use the support count or support percentage property. The default setting is **Percent**.
- **Support Count** specifies a minimum level of support to claim that items are associated (that is, they occur together in the database).
- **Support Percentage** specifies a minimum level of support to claim that items are associated (that is, they occur together in the database). The default frequency is 5%. The support percentage figure that you specify refers to the proportion of the largest single item frequency, and not the end support.
- **Maximum Items** determines the maximum size of the item set to be considered. For example, the default of four items indicates that a maximum of four items will be included in a single association rule.

If you are interested in associations that involve fairly rare products, you should consider reducing the support count or percentage when you run the Association node. If you obtain too many rules to be practically useful, you should consider raising the minimum support count or percentage as one possible solution.

Because you first want to perform an association analysis, you do not need the sequence variable.

11. Access the Variables dialog box for the Association node.

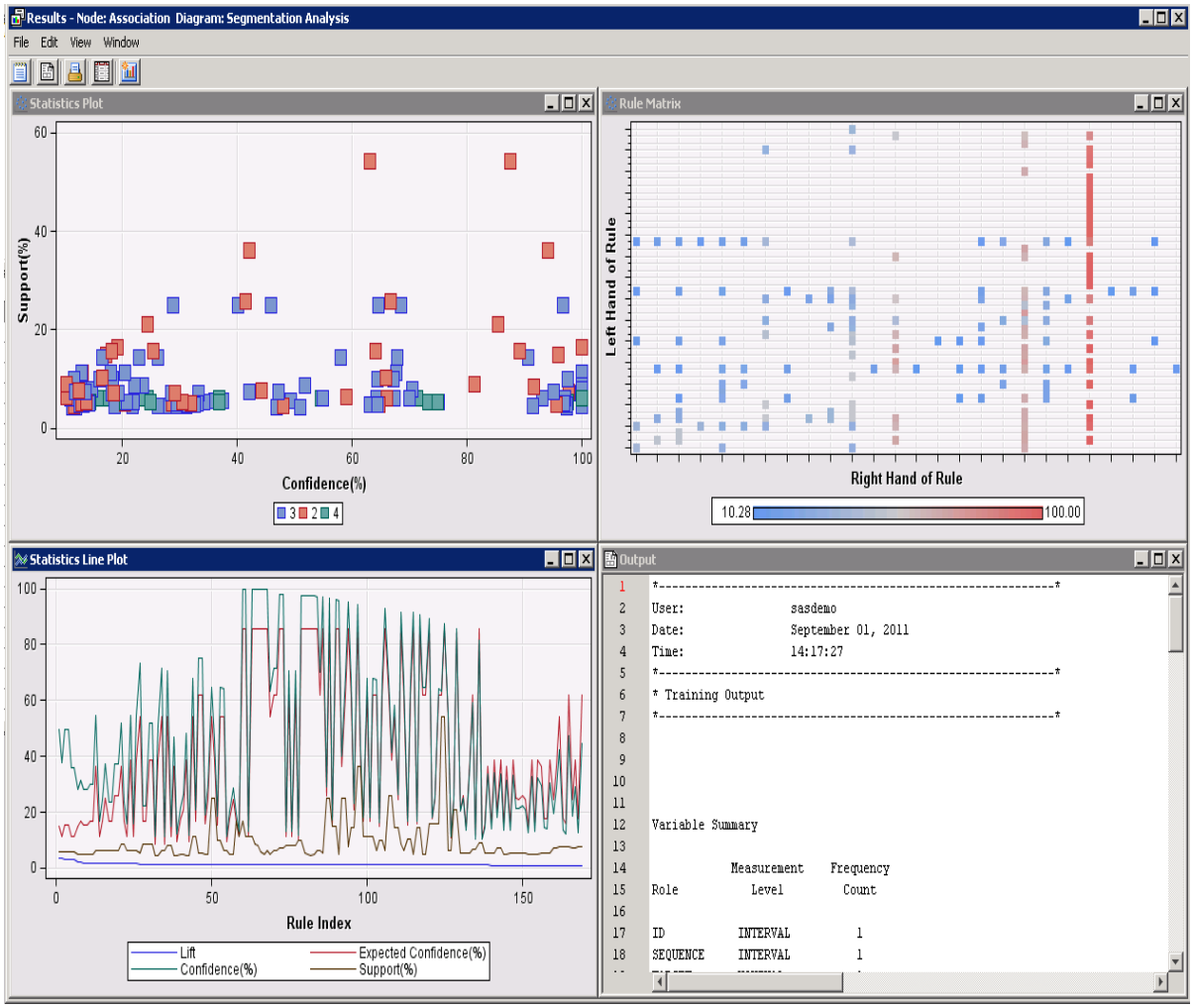
12. Select **Use** ⇨ **No** for the **VISIT** variable.



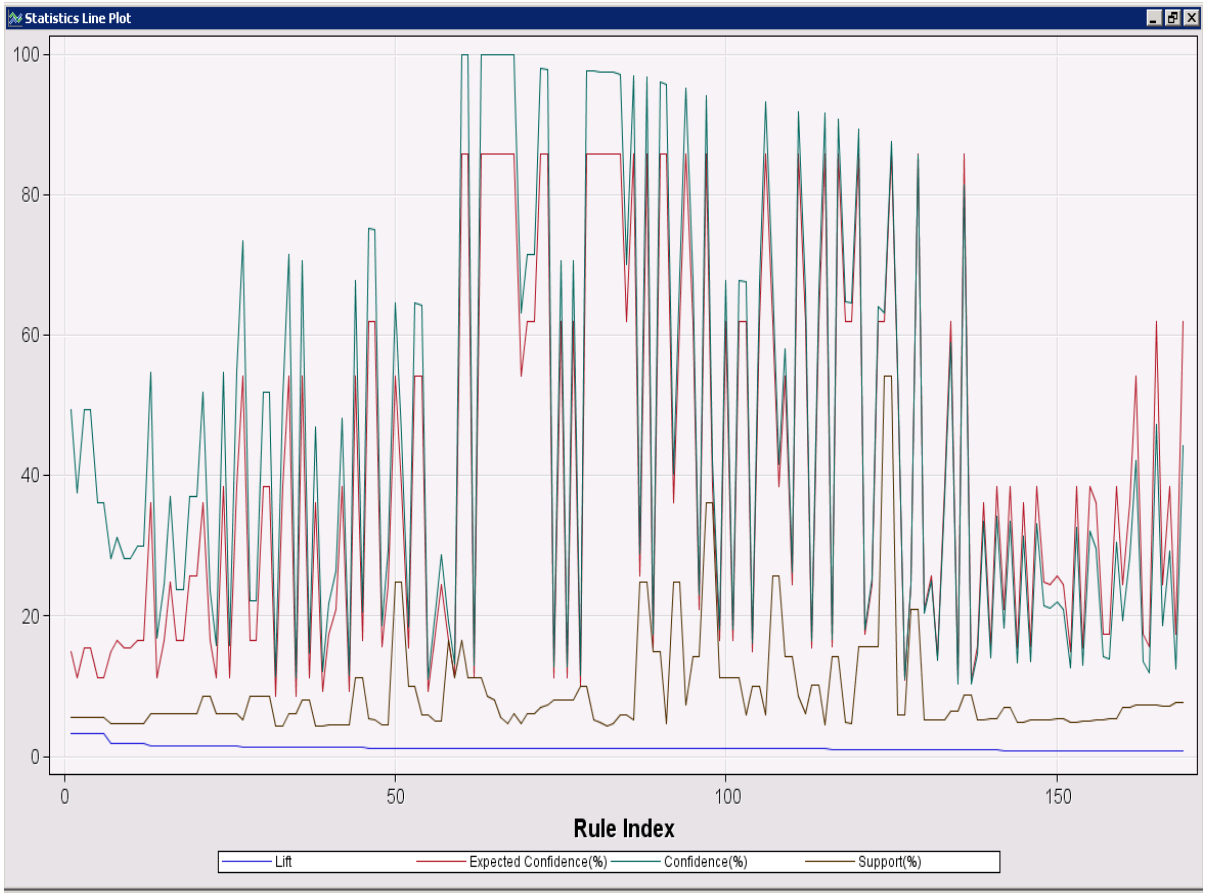
13. Select **OK** to close the Variables dialog box.

14. Run the diagram from the Association node and view the results.

The Results - Node: Association Diagram window appears with the Statistics Plot, Statistics Line Plot, Rule Matrix, and Output windows visible.



15. Maximize the Statistics Line Plot window.



The statistics line plot graphs the lift, expected confidence, confidence, and support for each of the rules by rule index number.

Consider the rule A ⇒ B. Recall the following:

- ***Support*** of A ⇒ B is the probability that a customer has both A and B.
- ***Confidence*** of A ⇒ B is the probability that a customer has B given that the customer has A.
- ***Expected Confidence*** of A ⇒ B is the probability that a customer has B.
- ***Lift*** of A ⇒ B is a measure of the strength of the association. If Lift=2 for the rule A=>B, then a customer having A is twice as likely to have B than a customer chosen at random. Lift is the confidence divided by the expected confidence.

Notice that the rules are ordered in descending order of lift.

16. To view the descriptions of the rules, select **View** ⇨ **Rules** ⇨ **Rule description**.

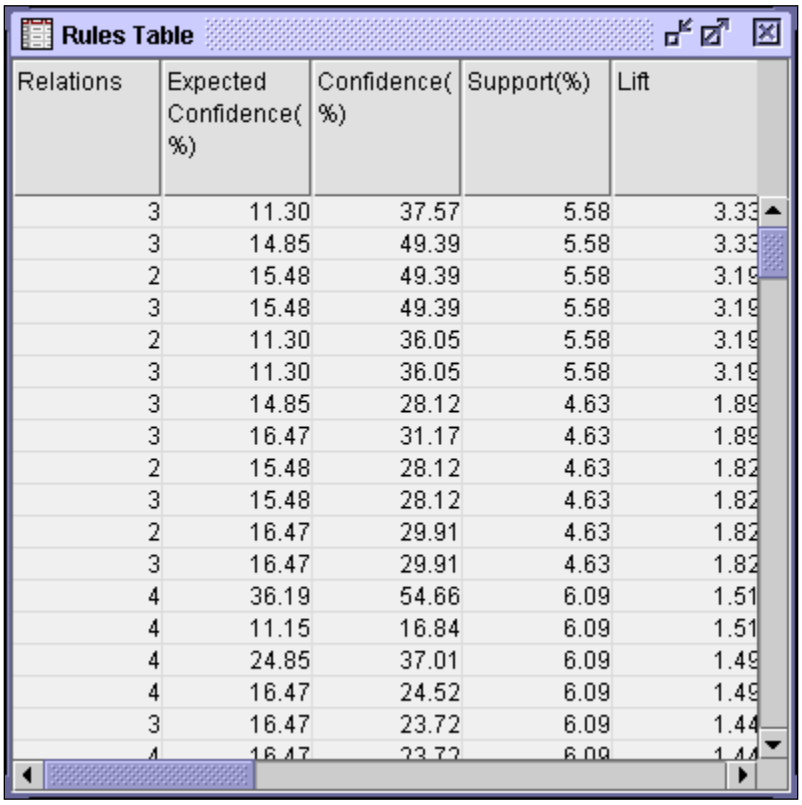
Rule description

MAP	RULE
RULE1	CKING & CCRD ==> CKCRD
RULE2	CKCRD ==> CKING & CCRD
RULE3	CKCRD ==> CCRD
RULE4	CKING & CKCRD ==> CCRD
RULE5	CCRD ==> CKCRD
RULE6	CCRD ==> CKING & CKCRD
RULE7	HMEQLC ==> CKING & CCRD
RULE8	CKING & CCRD ==> HMEQLC
RULE9	HMEQLC ==> CCRD
RULE10	HMEQLC & CKING ==> CCRD
RULE11	CCRD ==> HMEQLC
RULE12	CCRD ==> HMEQLC & CKING
RULE13	SVG & HMEQLC ==> CKING & ATM
RULE14	CKING & ATM ==> SVG & HMEQLC
RULE15	HMEQLC ==> SVG & CKING & ATM
RULE16	SVG & CKING & ATM ==> HMEQLC
RULE17	SVG & ATM ==> HMEQLC
RULE18	SVG & ATM ==> HMEQLC & CKING
RULE19	HMEQLC ==> SVG & ATM
RULE20	HMEQLC & CKING ==> SVG & ATM
RULE21	HMEQLC ==> CKING & ATM
RULE22	CKING & ATM ==> HMEQLC
RULE23	SVG & HMEQLC ==> ATM
RULE24	SVG & HMEQLC & CKING ==> ATM
RULE25	ATM ==> SVG & HMEQLC
RULE26	ATM ==> SVG & HMEQLC & CKING
RULE27	CD & ATM ==> SVG & CKING
RULE28	HMEQLC ==> ATM
RULE29	HMEQLC & CKING ==> ATM
RULE30	ATM ==> HMEQLC
RULE31	ATM ==> HMEQLC & CKING
RULE32	CKING & AUTO ==> ATM

The highest lift rule is checking, and credit card implies check card. This is not surprising given that many check cards include credit card logos. Notice the symmetry in rules 1 and 2. This is not accidental because, as noted earlier, lift is symmetric.

Another way to explore the rules found in the analysis is by plotting the Rules table.

17. Select **View** ⇨ **Rules** ⇨ **Rules Table**. The Rules Table window appears.



Rules Table

Relations	Expected Confidence(%)	Confidence(%)	Support(%)	Lift
3	11.30	37.57	5.58	3.33
3	14.85	49.39	5.58	3.33
2	15.48	49.39	5.58	3.19
3	15.48	49.39	5.58	3.19
2	11.30	36.05	5.58	3.19
3	11.30	36.05	5.58	3.19
3	14.85	28.12	4.63	1.89
3	16.47	31.17	4.63	1.89
2	15.48	28.12	4.63	1.82
3	15.48	28.12	4.63	1.82
2	16.47	29.91	4.63	1.82
3	16.47	29.91	4.63	1.82
4	36.19	54.66	6.09	1.51
4	11.15	16.84	6.09	1.51
4	24.85	37.01	6.09	1.49
4	16.47	24.52	6.09	1.49
3	16.47	23.72	6.09	1.44
4	16.47	23.72	6.09	1.44

The Rules Table contains all the information (confidence, support, etc.) on the rules generated by the analysis.

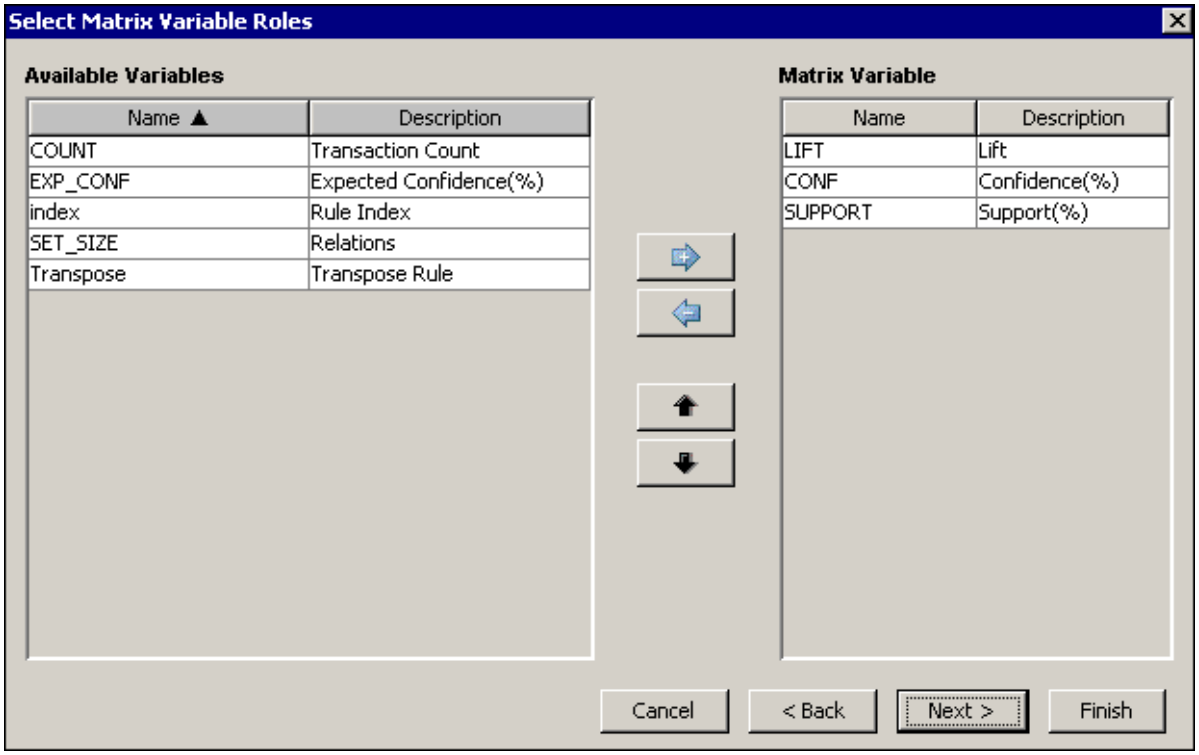
The values in a column can be sorted by clicking on the column header. This allows us to do the following:

- Given the customer has some existing product, what is the next best offer? We can do this by sorting according to an item on the LHS.
- If the bank is pushing a certain product, we can identify which product holders to offer it to. Do this by sorting according to the RHS.

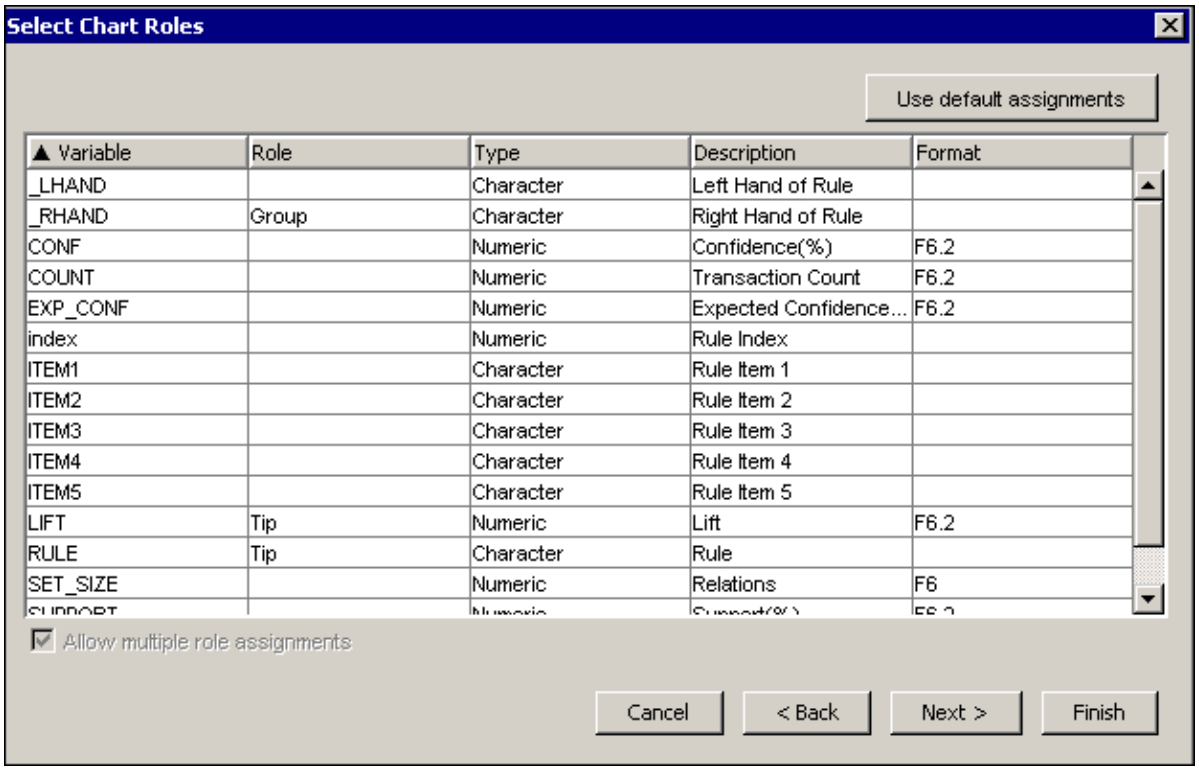
18. Select (the Plot Wizard icon).

19. Choose a Matrix graph for the type of chart, and select **Next >**.

20. Select the matrix variables: **Lift, Conf** and **Support** as shown below right. Select **Next** .

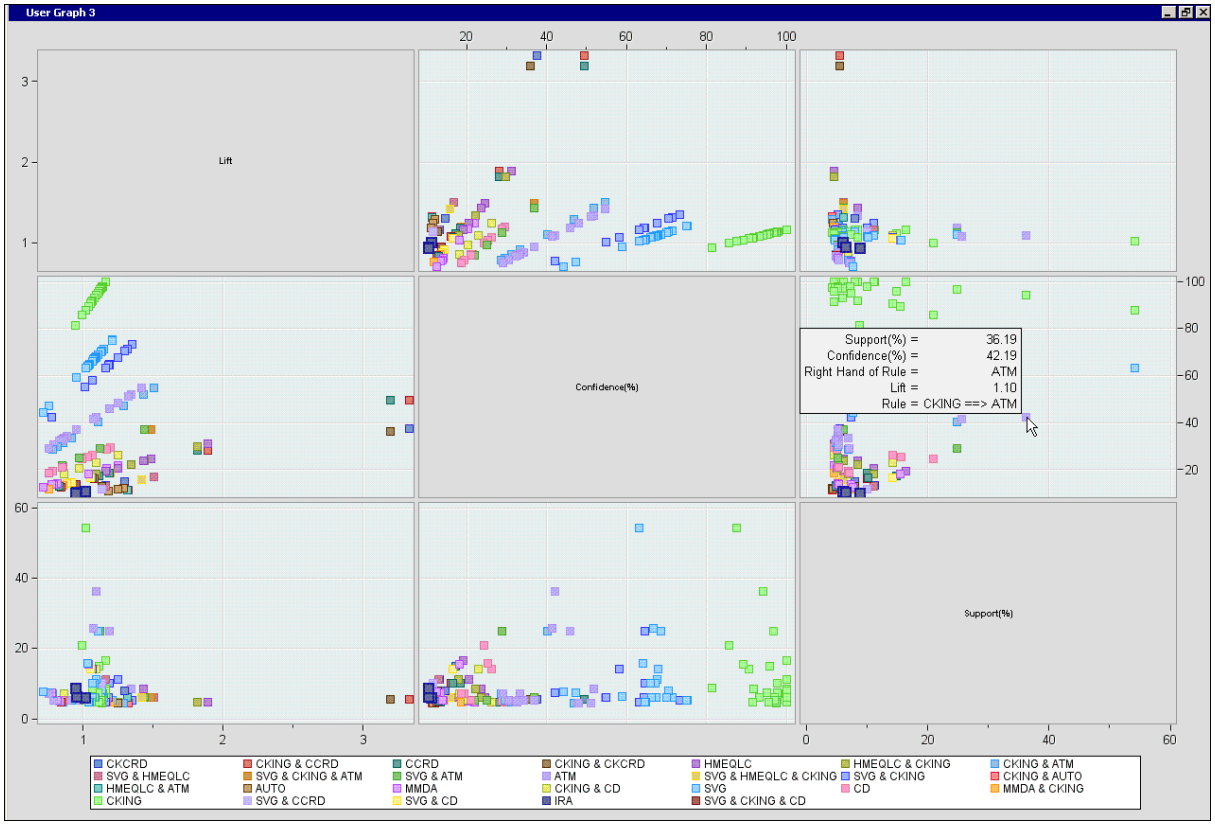


21. Select the **Group** role for **_RHAND** and the **Tip** role for **LIFT** and **RULE** to add these details to the tooltip action.



Variable	Role	Type	Description	Format
_LHAND		Character	Left Hand of Rule	
_RHAND	Group	Character	Right Hand of Rule	
CONF		Numeric	Confidence(%)	F6.2
COUNT		Numeric	Transaction Count	F6.2
EXP_CONF		Numeric	Expected Confidence...	F6.2
index		Numeric	Rule Index	
ITEM1		Character	Rule Item 1	
ITEM2		Character	Rule Item 2	
ITEM3		Character	Rule Item 3	
ITEM4		Character	Rule Item 4	
ITEM5		Character	Rule Item 5	
LIFT	Tip	Numeric	Lift	F6.2
RULE	Tip	Character	Rule	
SET_SIZE		Numeric	Relations	F6

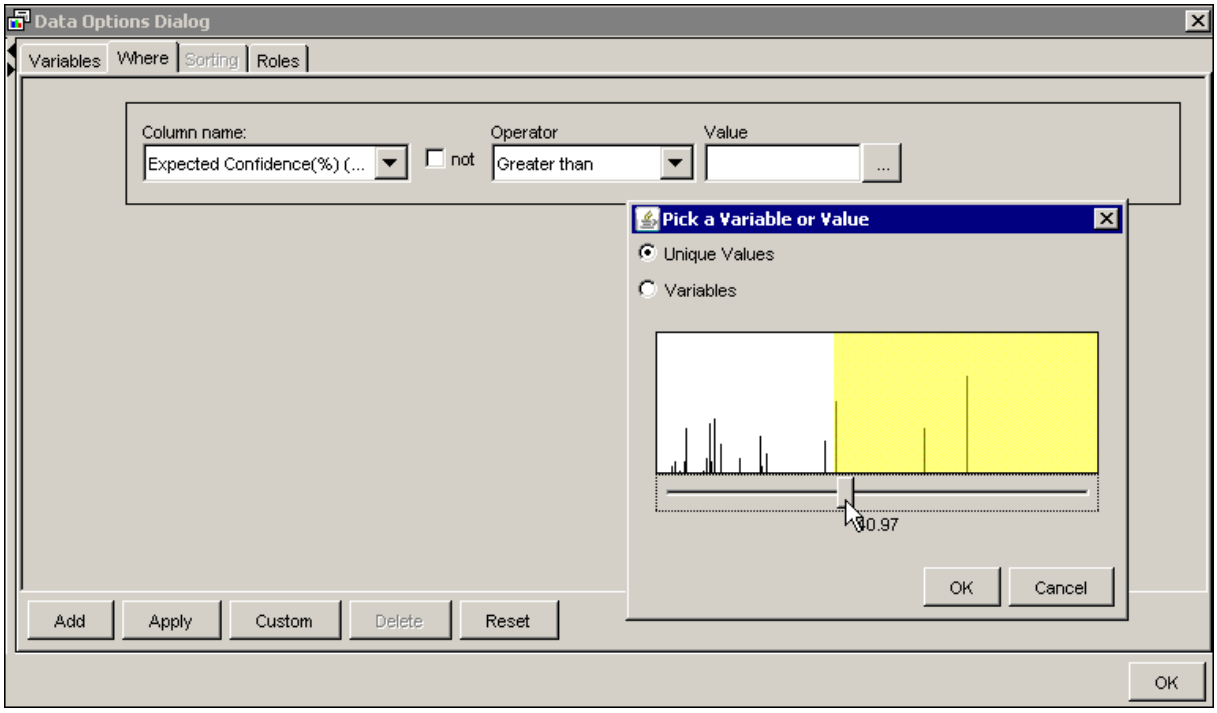
22. Select **Finish** to generate the plot.



The legend shows the right hand of the rule. When you click a service or group of services in the legend, the points in the matrix graphs are highlighted. This plot enables you to explore the relationships among the various metrics in association analysis.

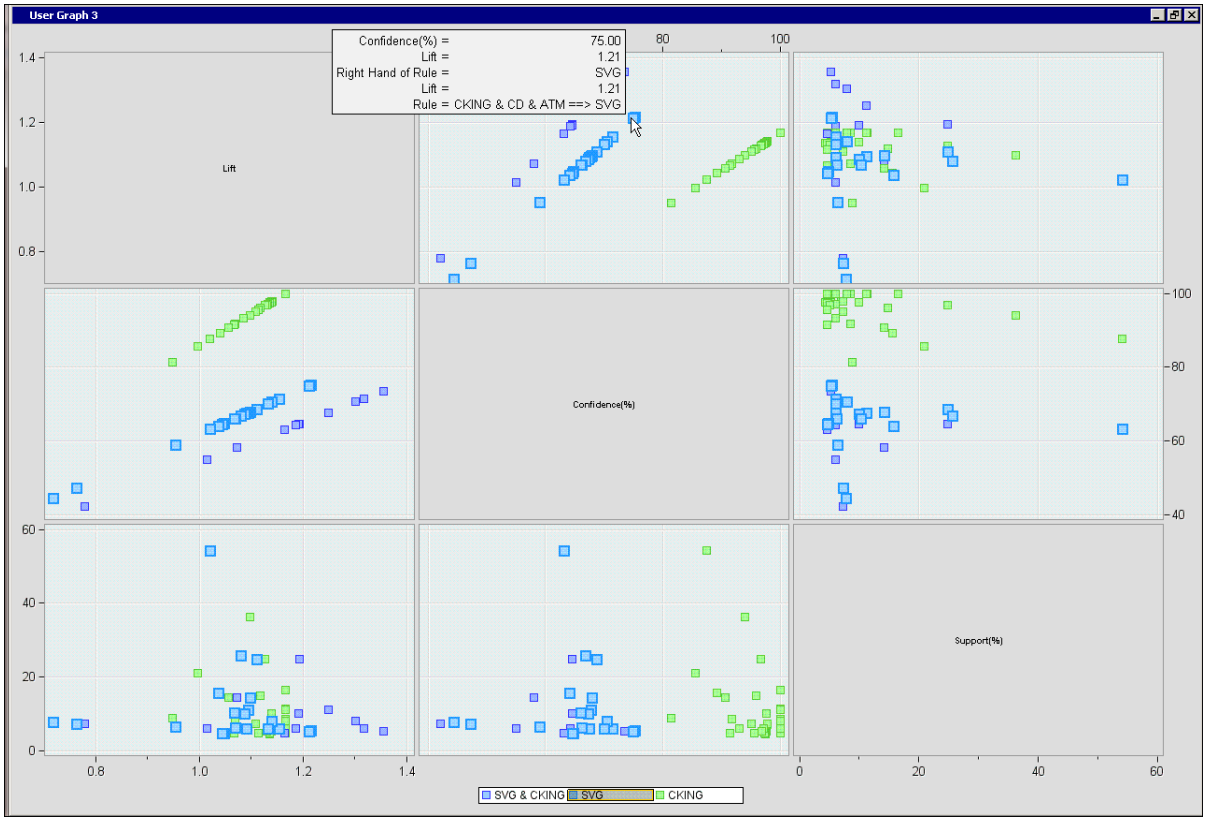
When you hover the cursor over a selected point in the plot, the tooltips show the details of the point, including the full rule.

23. Right-click in the graph and select **Data Options**. Select the **Where** tab. Specify **Expected Confidence(%)** as the column name and **Greater than** as the operator. Click the elipses next to **Value**. Set the slider to include values greater than 40, or type **40** for the value.

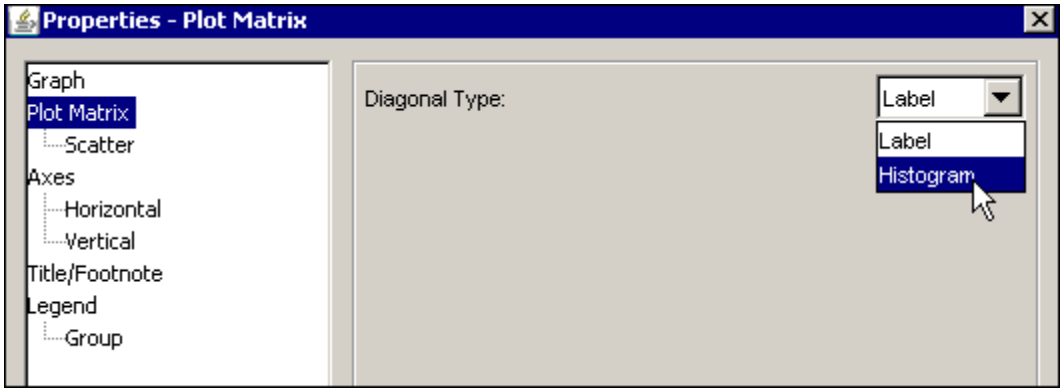


24. Select **OK** and **Apply**.

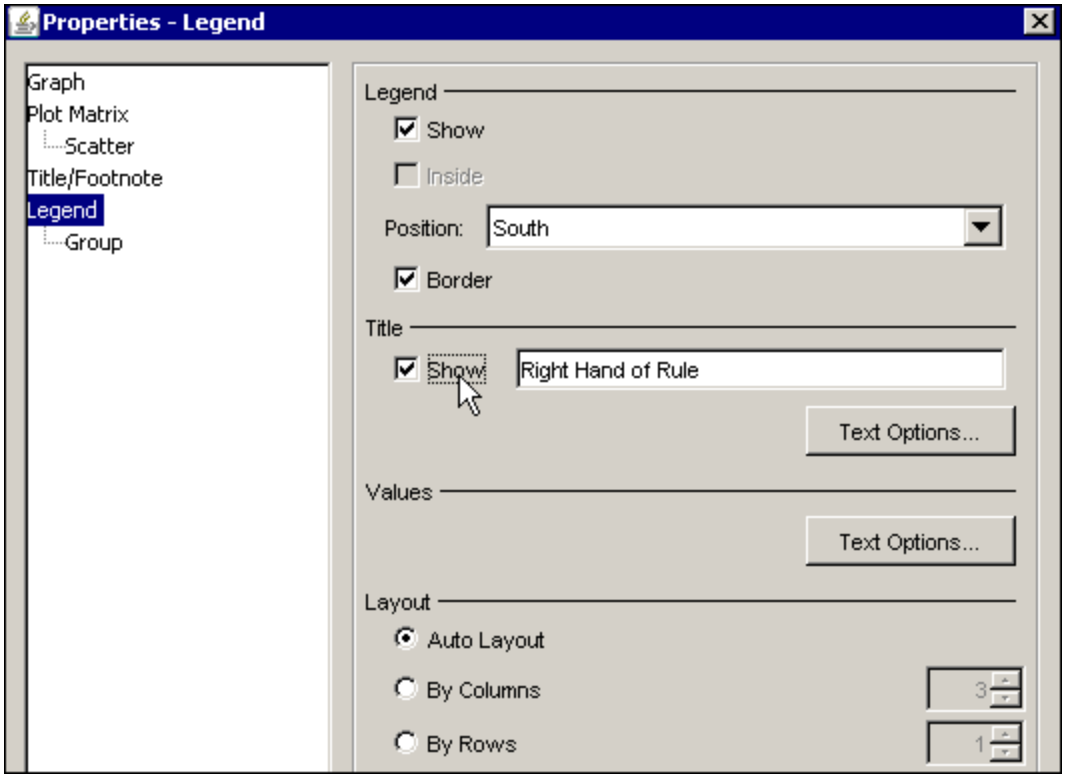
25. Select **OK** . The subset selected cases represent three different sets of services in the legend for the right hand of the rules.



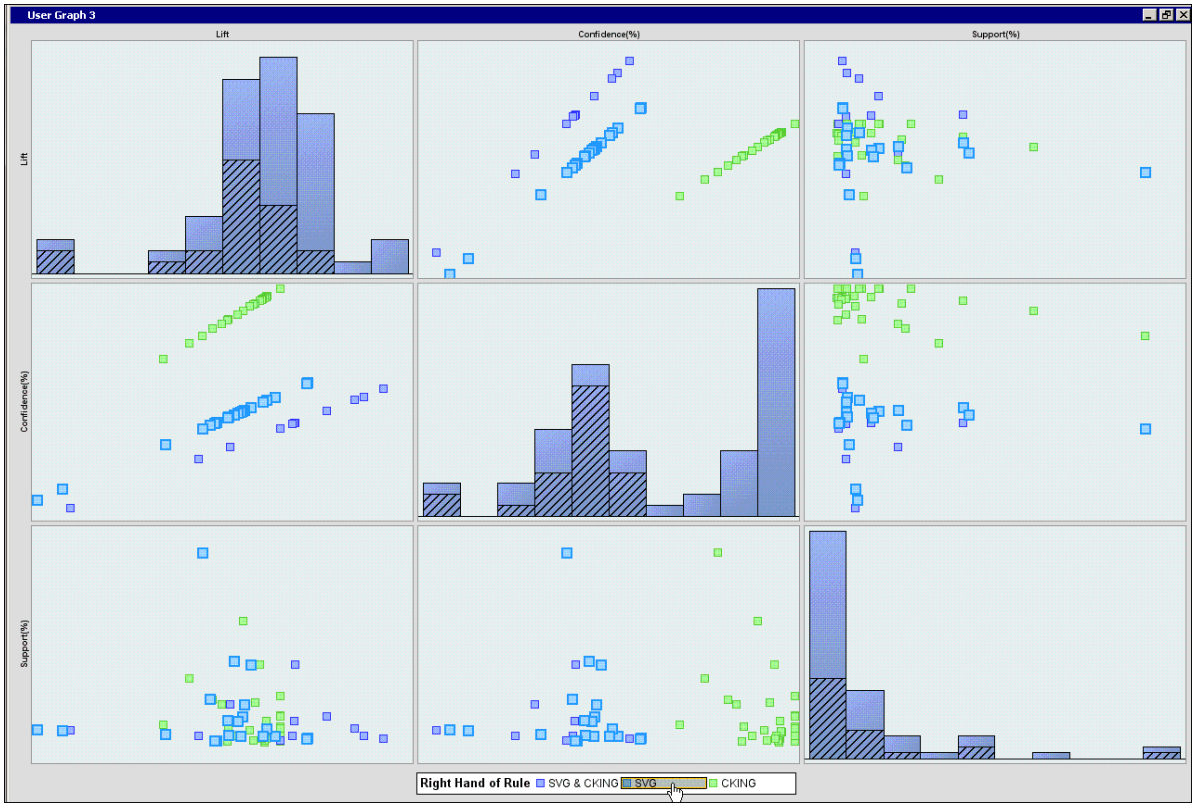
26. You can modify the look of the graph. Right-click the graph and select **Graph Properties**. Change the plot matrix diagonal type to **Histogram**.



27. Label the legend by selecting **Legend** and selecting the check box in the Title are next to **Show (Right hand of Rule)**. Select **OK**.



28. Click the **SVG** (Savings Account) category in the legend and notice that the histograms show the distribution of the selected rules in the diagonal.



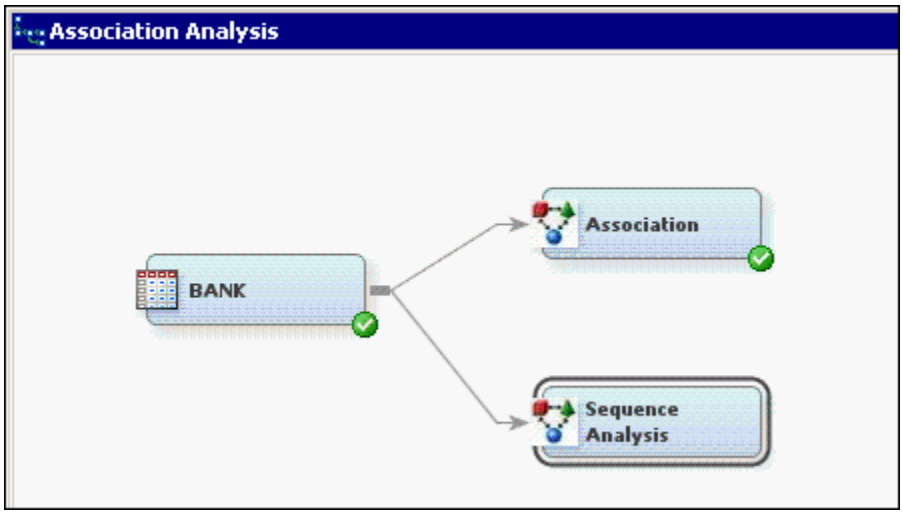
29. Close the Results window.

Sequence Analysis Example

In addition to the products owned by its customers, the bank is interested in examining the order in which the products are purchased. The sequence variable in the data set enables you to conduct a sequence analysis.

1. Add an **Association** node to the diagram workspace and connect it to the **BANK** node.

2. Rename the new node **Sequence Analysis**.



3. Examine the Sequence panel in the Properties panel.

Sequence	
Chain Count	3
Consolidate Time	0.0
Maximum Transaction Duration	.
Support Type	Percent
Support Count	.
Support Percentage	2.0

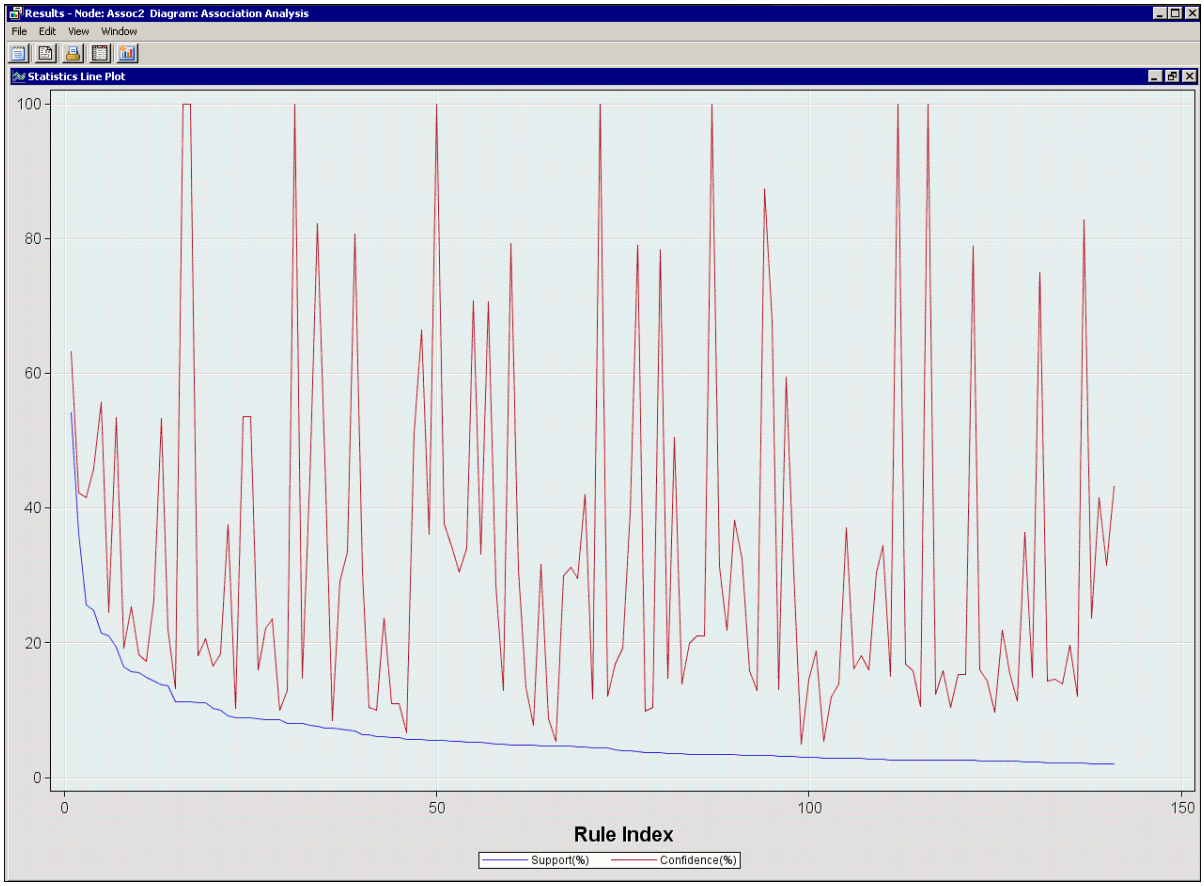
The options in the Sequence panel enable you to specify the following properties:

- **Chain Count** is the maximum number of items that can be included in a sequence. The default value is 3 and the maximum value is 10.
- **Consolidate Time** enables you to specify whether consecutive visits to a location or consecutive purchases over a given interval can be consolidated into a single visit for analysis purposes. For example, two products purchased less than a day apart might be considered to be a single transaction.

- **Maximum Transaction Duration** enables you to specify the maximum length of time for a series of transactions to be considered a sequence. For example, you might want to specify that the purchase of two products more than three months apart does not constitute a sequence.
- **Support Type** specifies whether the sequence analysis should use the Support Count or Support Percentage property. The default setting is **Percent**.
- **Support Count** specifies the minimum frequency required to include a sequence in the sequence analysis when the Sequence Support Type property is set to **Count**. If a sequence has a count less than the specified value, that sequence is excluded from the output.
- **Support Percentage** specifies the minimum level of support to include the sequence in the analysis when the Support Type property is set to **Percent**. If a sequence has a frequency that is less than the specified percentage of the total number of transactions, then that sequence is excluded from the output. The default percentage is 2%. Permissible values are real numbers between 0 and 100.

4. Run the diagram from the Sequence Analysis node and view the results.

5. Maximize the Statistics Line Plot window.



The statistics line plot graphs the confidence and support for each of the rules by rule index number.

The *percent support* is the transaction count divided by the total number of customers, which would be the maximum transaction count. The *percent confidence* is the transaction count divided by the transaction count for the left side of the sequence.

6. Select **View** ⇨ **Rules** ⇨ **Rules Table** to view the details of the rules.

Results - Node: Sequence Analysis Diagram: Association Analysis

File Edit View Window

Rules Table

Chain Length	Transaction Count	Support(%)	Confidence(%)	Rule	Chain Item 1	Chain Item 2	Chain Item 3	Rule Index	Left Hand of Rule	Right Hand of Rule
2	4329	54.17	63.15	CKING ==> ...	CKING	SVG		1	CKING	SVG
2	2892	36.19	42.19	CKING ==> ...	CKING	ATM		2	CKING	ATM
2	2053	25.69	41.53	SVG ==> ATM	SVG	ATM		3	SVG	ATM
3	1986	24.85	45.88	CKING ==> ...	CKING	SVG	ATM	4	CKING ==> ...	ATM
2	1709	21.39	55.61	ATM ==> ATM	ATM	ATM		5	ATM	ATM
2	1677	20.99	24.46	CKING ==> ...	CKING	CD		6	CKING	CD
3	1546	19.35	53.46	CKING ==> ...	CKING	ATM	ATM	7	CKING ==> ...	ATM
2	1316	16.47	19.20	CKING ==> ...	CKING	HMEQLC		8	CKING	HMEQLC
2	1256	15.72	25.40	SVG ==> CD	SVG	CD		9	SVG	CD
2	1245	15.58	18.16	CKING ==> ...	CKING	MMDA		10	CKING	MMDA
2	1187	14.85	17.32	CKING ==> ...	CKING	CCRD		11	CKING	CCRD
3	1139	14.25	26.31	CKING ==> ...	CKING	SVG	CD	12	CKING ==> ...	CD
3	1094	13.69	53.29	SVG ==> AT...	SVG	ATM	ATM	13	SVG ==> ATM	ATM
2	1085	13.58	21.95	SVG ==> S...	SVG	SVG		14	SVG	SVG
2	903	11.30	100.00	CKCRD ==...	CKCRD	CKCRD		16	CKCRD	CKCRD
2	903	11.30	13.17	CKING ==> ...	CKING	CKCRD		15	CKING	CKCRD
3	903	11.30	100.00	CKING ==> ...	CKING	CKCRD	CKCRD	17	CKING ==> ...	CKCRD
2	891	11.15	18.02	SVG ==> H...	SVG	HMEQLC		18	SVG	HMEQLC
3	891	11.15	20.58	CKING ==> ...	CKING	SVG	HMEQLC	19	CKING ==> ...	HMEQLC
2	817	10.22	16.53	SVG ==> C...	SVG	CCRD		20	SVG	CCRD
3	798	9.99	18.43	CKING ==> ...	CKING	SVG	CCRD	21	CKING ==> ...	CCRD
2	736	9.21	37.55	CD ==> CD	CD	CD		22	CD	CD
2	705	8.82	10.28	CKING ==> ...	CKING	IRA		23	CKING	IRA
2	704	8.81	53.50	HMEQLC =...	HMEQLC	HMEQLC		24	HMEQLC	HMEQLC

In newer versions of SAS Enterprise Miner, a Pseudo-lift is computed for the rules.